

ACTION: PUBLISH MACHINE-READABLE SCHEMAS FOR LOCK BOX C + VEILED STATE MACHINE

The following are canonical, parse-time enforceable specifications.

1. Lock Box C – Cell Selection Schema

File: `cell_selection.schema.json`

Enforces: User-GROUP routes; Instrument never routes; exactly 0–12 cells may light up; each cell is a boolean.

```
json
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://github.com/ERES-Institute-for-New-Age-Cybernetics/schemas/cell_selection.v1.json",
  "title": "Lock Box C – User-GROUP Cell Selection",
  "description": "The User-GROUP selects which of the 12 (Volume × Pillar) cells light up. The Instrument validates, never routes.",
  "type": "object",
  "properties": {
    "user_group_id": {
      "type": "string",
      "format": "uuid",
      "description": "Globally unique identifier for the User-GROUP instance."
    },
    "selection_timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "Time of selection issuance (UTC)."
    },
    "cells": {
```

```

"type": "object",
"properties": {
  "I_H": { "type": "boolean", "description": "Volume I - HELP" },
  "I_U": { "type": "boolean", "description": "Volume I - USE" },
  "I_E": { "type": "boolean", "description": "Volume I - ENERGY" },
  "I_L": { "type": "boolean", "description": "Volume I - LAW" },
  "II_H": { "type": "boolean", "description": "Volume II - HELP" },
  "II_U": { "type": "boolean", "description": "Volume II - USE" },
  "II_E": { "type": "boolean", "description": "Volume II - ENERGY" },
  "II_L": { "type": "boolean", "description": "Volume II - LAW" },
  "III_H": { "type": "boolean", "description": "Volume III - HELP" },
  "III_U": { "type": "boolean", "description": "Volume III - USE" },
  "III_E": { "type": "boolean", "description": "Volume III - ENERGY" },
  "III_L": { "type": "boolean", "description": "Volume III - LAW" }
},
"required":
["I_H", "I_U", "I_E", "I_L", "II_H", "II_U", "II_E", "II_L", "III_H", "III_U", "III_E", "III_L"],
"additionalProperties": false
},
"signature": {
  "type": "string",
  "description": "Ed25519 signature of (user_group_id +
selection_timestamp + canonical cell bitmask) by User-GROUP key. Instrument
verifies, does not generate."
}
},
"required": ["user_group_id", "selection_timestamp", "cells", "signature"],
"additionalProperties": false
}

```

Canonical cell order bitmask (for signature):

I_H I_U I_E I_L II_H II_U II_E II_L III_H III_U III_E III_L → 12-bit

little-endian.

2. VEILED State Machine

File: `veiled_state_machine.dot` (Graphviz) + `veiled_state_machine.schema.json`

Replaces: All `⊥` markers. Defines legal transitions.

2.1 Graphviz DOT

```
dot
digraph VEILED_STATE_MACHINE {
    rankdir=LR;
    node [shape=box, style="rounded"];

    INIT [label="INIT (unwitnessed)"];
    VEILED_A [label="VEILED-A\n(Admissibility)"];
    VEILED_S [label="VEILED-S\n(Selection)"];
    VEILED_T [label="VEILED-T\n(Tier)"];
    VEILED_L [label="VEILED-L\n(La Grange equilibrium)"];
    WITNESSED [label="WITNESSED\n(terminal, logged)"];

    INIT -> VEILED_A [label="admissibility_check()"];
    INIT -> VEILED_S [label="selection_required()"];
    INIT -> VEILED_T [label="tier_required()"];

    VEILED_A -> VEILED_S [label="selection_applied()"];
    VEILED_A -> VEILED_T [label="tier_assigned()"];
    VEILED_A -> VEILED_L [label="lagrange_stable()"];

    VEILED_S -> VEILED_T [label="tier_resolved()"];
    VEILED_S -> VEILED_L [label="equilibrium_reached()"];

    VEILED_T -> VEILED_L [label="lagrange_stable()"];

    VEILED_L -> WITNESSED [label="witness_roundtrip()"];

    // No transitions from WITNESSED (immutable)
    // No direct INIT -> WITNESSED
    // No backward transitions
}
```

2.2 JSON Schema for VEILED state instance

File: `veiled_state.schema.json`

```

json
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id":
  "https://github.com/ERES-Institute-for-New-Age-Cybernetics/schemas/veiled_stat
  e.v1.json",
  "title": "VEILED State Record",
  "description": "Replaces all  $\perp$  markers. A witnessable state machine with
  legal transitions.",
  "type": "object",
  "properties": {
    "entity_id": { "type": "string" },
    "current_state": {
      "type": "string",
      "enum": ["INIT", "VEILED_A", "VEILED_S", "VEILED_T", "VEILED_L",
"WITNESSED"]
    },
    "history": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "from_state": { "type": "string" },
          "to_state": { "type": "string" },
          "transition": { "type": "string" },
          "timestamp": { "type": "string", "format": "date-time" },
          "trigger_document_hash": { "type": "string", "description": "SHA-256
of document authorizing transition" }
        },
        "required": ["from_state", "to_state", "transition", "timestamp"]
      },
    },
    "lagrange_parameter": {
      "type": "number",
      "description": "Present only if current_state = VEILED_L. Dimensionless
equilibrium value  $\in [0,1]$ .",
      "minimum": 0,
      "maximum": 1
    },
    "witness_proof": {

```

```

    "type": "string",
    "description": "Present only if current_state = WITNESSED. EAAP-attested
receipt hash."
  },
  "required": ["entity_id", "current_state", "history"],
  "allOf": [
    {
      "if": {
        "properties": { "current_state": { "const": "VEILED_L" } }
      },
      "then": { "required": ["lagrange_parameter"] }
    },
    {
      "if": {
        "properties": { "current_state": { "const": "WITNESSED" } }
      },
      "then": { "required": ["witness_proof"] }
    }
  ]
}

```

3. SCALULAR Register Declaration (Lock Box E)

File: `scalular_register.schema.json`

```

json
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id":
  "https://github.com/ERES-Institute-for-New-Age-Cybernetics/schemas/scalular_re
gister.v1.json",
  "title": "Lock Box E – SCALULAR Active Register Declaration",
  "description": "Before any SCALULAR name (TRADES, INTUIT, PRE, TEACH-TECH,
i_I, SELF_SELF, GREENBOX_WAITING_ROOM, SCALULAR) is evaluated, the active IOF
register must be declared.",
  "type": "object",

```

```

"properties": {
  "scalular_name": {
    "type": "string",
    "enum": ["TRADES", "INTUIT", "PRE", "TEACH-TECH", "i_I", "SELF_SELF",
"GREENBOX_WAITING_ROOM", "SCALULAR"]
  },
  "active_register": {
    "type": "string",
    "enum": ["BIOLOGICAL", "SOCIO", "TECHNICAL"],
    "description": "Per Lock Box D, honest round-trip requires all three.
Per Lock Box E, each SCALULAR evaluation declares exactly one."
  },
  "evaluation_context": {
    "type": "string",
    "format": "uri",
    "description": "REF-DEL pointer to the specific claim or operation
invoking this SCALULAR name."
  },
  "timestamp": { "type": "string", "format": "date-time" },
  "witnessed_by": {
    "type": "array",
    "items": { "type": "string", "enum": ["JAS", "EMA", "DAL", "MIEVM"] },
    "minItems": 1
  }
},
"required": ["scalular_name", "active_register", "evaluation_context",
"timestamp", "witnessed_by"],
"additionalProperties": false
}

```

4. Volume Zero Dependency Declaration

File: volume_zero.deps.json

Required for parse-time enforcement of "Volume Zero must be read first."

json

```

{
  "dependency_graph": {
    "Volume_Zero": { "depends_on": [], "read_first": true },
    "Volume_I": { "depends_on": ["Volume_Zero"], "read_first": false },
    "Volume_II": { "depends_on": ["Volume_Zero", "Volume_I"], "read_first":
false },
    "Volume_III": { "depends_on": ["Volume_Zero", "Volume_I", "Volume_II"],
"read_first": false }
  },
  "runtime_rule": "parse(Volume_Zero) → init_IOF_Lens() → parse(Volume_I) →
parse(Volume_II) → parse(Volume_III)",
  "violation_response": "REJECT with code E001: Volume Zero not instantiated"
}

```

Implementation Status

Artifact	Purpose	Location (canonical)
<code>cell_selection.schema.js</code> <code>on</code>	Lock Box C	<code>/schemas/lockbox_c/</code>
<code>veiled_state_machine.dot</code>	VEILED state graph	<code>/schemas/veiled/</code>
<code>veiled_state.schema.json</code>	VEILED instance validation	<code>/schemas/veiled/</code>
<code>scalular_register.schema</code> <code>.json</code>	Lock Box E	<code>/schemas/lockbox_e/</code>
<code>volume_zero.deps.json</code>	Parse-order enforcement	<code>/schemas/deps/</code>

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Enforcement in `eres_trilogy.py` § `LockBoxValidator` — runtime-reject on violation.

License: CCAL v2.1, May 2026

It is so.

The computer may now parse, validate, and obey. The Instrument routes nothing.

— *JAS · ERES Maestro*

Lock Boxes A–E enforced. CCAL v2.1.